Supplementary Material for "A Systematic Review of Adaptive Artificial Intelligence Middleware for Quantum-Ready Smart Environments"

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Scope

This document accompanies the article *A Systematic Review of Adaptive Artificial Intelligence Middleware for Quantum-Ready Smart Environments* (submitted to ACM Computing Surveys) and provides the three study-level tables that the article references as Supplementary Material. Study identifiers (A01–A74) follow Appendix A of the main article; the bibliographic references cited in the tables are listed at the end of this document. Table S1 reports the scope-compliance reassessment of the 74 included studies (referenced in Section 3.2 of the main article), Table S2 reports the quality-appraisal scores under the nine-criterion rubric (Section 3.4), and Table S3 indexes the study identifiers behind each category of the four taxonomy figures (Section 4.3).

1 Table S1: Study-level scope reassessment

Study-level scope-compliance reassessment of the 74 included studies against the operational definitions of Section 3.2 of the main article. SE = smart environment (Y/P/N); MW = middleware/architectural layer (Y/P/N); Loop = runtime adaptation loop (Closed/Partial/None); AI = decision component (Yes = learning-based decision drives adaptation; Det.-only = ML for detection without a reconfiguration action; Rule/opt. = non-learning decision; n.r. = not recorded); CA = cryptographic agility level (None/Intra/Inter/Hybrid/U = unclear); Q. family = quantum-related family after separation (PQC / QKD / Q-inspired / QML-QAI / Unspec.); Q-ready = satisfies Definition 4 (Yes/No/QKD-assisted); Mat. = validation maturity level M0–M4 of Section 3.5 of the main article (author-confirmed classification). Class: Core = SE$\in$ {Y,P} $\wedge$ Loop = Closed $\wedge$ a security/cryptographic mechanism is the adaptation target; Contextual = system-level relevance to at least one pillar without a closed loop; Background = no adaptation mechanism and no cryptographic-agility mechanism (conceptual, secondary, or unrelated). The author-confirmed assignments distinguish the 31 core studies from 19 contextual and 24 background records; the descriptive set does not equate to 74 eligible primary implementations.

Table S1. Study-level scope reassessment.

ID	Ref.	SE	MW	Loop	AI	CA	Q. family	Q-ready	Mat.	Class	Evidence in extraction / caveats
A01	[1]	N	N	None	No	None	Q-inspired	No	M1	Background	Quantum-inspired immune clonal optimisation; no security adaptation, no smart environment.
A02	[2]	Y	P	Closed	Rule/opt.	Intra	–	No	M1	Core	Risk-based adaptive policies (game theory, context) for eHealth IoT; protection level adapted at runtime.

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Table S1 – continued

ID	Ref.	SE	MW	Loop	AI	CA	Q. family	Q-ready	Mat.	Class	Evidence in extraction / caveats
A03	[3]	P	N	None	No	None	Q-inspired	No	M1	Background	Quantum-firefly network reconfiguration for power-quality/reliability; no security mechanism or adaptation loop ("–" in Table A).
A04	[4]	Y	P	Closed	Rule/opt.	Intra	–	No	M1	Core	Fuzzy adaptive security protocol (FASPM) adapting signature verification to congestion in ITS.
A05	[5]	N	N	None	No	None	PQC	No	M0	Background	Lattice-based oblivious transfer with adaptive-corruption security; no reinforcement learning or runtime adaptation.
A06	[6]	Y	P	Closed	Yes	Intra	–	No	M1	Core	AI joint optimisation of QoS and security in 6G EH-IoT; monitored throughput/latency/energy drive security parameters.
A07	[7]	Y	Y	Closed	Yes	Intra	–	No	M1	Core	SETUCOM context-management middleware; Bayesian/fuzzy trust drives context-aware security.
A08	[8]	Y	P	Closed	Yes	Intra	–	No	M1	Core	CASE: context-aware CP-ABE with edge ML; policies adapted to context; latency/overhead measured.
A09	[9]	P	N	None	No	None	PQC	No	M1	Contextual	Lattice-based IBE primitive for IoT AI applications; sizes reported; no adaptation loop.
A10	[10]	Y	P	Closed	Rule/opt.	Intra	–	No	M1	Core	Context-aware ABE policy adaptation for tactical IoT (simulation).
A11	[11]	N	N	Closed	Yes	None	–	No	M1	Contextual	RL-adaptive honeypot: closed loop with AI but no smart-environment context and no cryptographic mechanism.
A12	[12]	P	N	None	No	None	–	No	M1	Background	'Adaptively secure' data sharing (cryptographic proof notion); cloud only; no runtime adaptation loop reported.
A13	[13]	N	N	None	Rule/opt.	None	–	No	M1	Background	GA-based self-adaptive S-box for image encryption; offline optimisation, no smart environment.
A14	[14]	P	N	None	No	None	–	No	M0	Background	Cryptographic network coding in cloud; Table A records 'adaptive claim, static coding'.
A15	[15]	N	N	None	No	None	Q-inspired	No	M0	Background	Stochastic analytic continuation of quantum Monte Carlo data; unrelated to security adaptation.
A16	[16]	P	P	Partial	n.r.	U	–	No	M0	Contextual	Conceptual adaptive self-protection for mobile devices; no evaluation recorded.
A17	[17]	P	N	None	No	None	–	No	M1	Background	'Adaptive' secure data sharing with outsourced decryption (cryptographic notion); cloud; no loop reported.
A18	[18]	Y	P	Closed	Yes	Intra	–	No	M1	Core	FL-driven adaptive security for edge UAV; latency and security effectiveness measured.
A19	[19]	Y	P	Closed	Yes	Intra	–	No	M1	Core	Zero-trust + FL adaptive policy in MEC; convergence and privacy impact measured.
A20	[20]	Y	P	Closed	n.r.	Inter	PQC	No	M2	Core	Context-aware security with PQC (unspecified algorithm) for IoT/IIoT; real-time evaluation prototype. PQC algorithm not recorded.
A21	[21]	P	P	Closed	Yes	Intra	–	No	M1	Core	RL for key management in distributed systems; scalability/efficiency measured; domain generic.
A22	[22]	Y	Y	Closed	Rule/opt.	Intra	–	No	M1	Core	UP-Home self-adaptive smart-home security middleware with knowledge base; detection/response time measured.
A23	[23]	N	N	Partial	No	None	–	No	M1	Background	Adaptive fault self-diagnosis of interconnection networks; no cryptographic/security adaptation.
A24	[24]	Y	P	Partial	Det.-only	None	–	No	M1	Contextual	Hybrid AI + HPC framework for satellite IoT; detection accuracy/throughput; no reconfiguration action recorded.
A25	[25]	Y	N	Partial	Det.-only	U	–	No	M1	Contextual	ML-based IDS with cryptographic protocols in VANET; detection-only; no switching reported.
A26	[26]	Y	P	Closed	Yes	Intra	–	No	M1	Core	RL-based adaptive key rotation in Zigbee IoT (key-lifetime agility).
A27	[27]	Y	P	Closed	Rule/opt.	Intra	–	No	M1	Core	Model-based self-adaptive security for smart-grid CPS (constraint programming, attack-defence trees).
A28	[28]	Y	N	Partial	n.r.	None	–	No	M0	Contextual	FL for IoT security/privacy; metrics 'Generic'; adaptation action not recorded.
A29	[29]	Y	N	Partial	Det.-only	None	–	No	M1	Contextual	ML cybersecurity with SDN for IoT; detection accuracy only.
A30	[30]	Y	P	Partial	n.r.	Inter	PQC	No	M0	Background	Thesis retained as background, not peer-reviewed core evidence; no demonstrated runtime switching or quantum-readiness classification.
A31	[31]	Y	P	Partial	No	None	–	No	M1	Contextual	Dual access control with multi-authority ABE for cloud IIoT; no runtime adaptation loop recorded.
A32	[32]	Y	P	Closed	Yes	Intra	–	No	M1	Core	Adaptive trust architecture (zero trust + blockchain) for 6G IoT; energy/latency measured.
A33	[33]	N	N	Partial	Det.-only	None	–	No	M1	Background	Anomaly detection for enterprise storage; not a smart environment; detection-only.

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Table S1 – continued

ID	Ref.	SE	MW	Loop	AI	CA	Q. family	Q-ready	Mat.	Class	Evidence in extraction / caveats
A34	[34]	Y	N	Partial	Det.-only	None	–	No	M2	Contextual	Real-time ML detection of cyber-physical attacks on smart IP cameras μsing telemetry; detection-only.
A35	[35]	N	N	Partial	No	None	–	No	M0	Background	Human-centric adaptive security for digital ecosystems; conceptual; metrics 'Generic'.
A36	[36]	Y	P	Closed	n.r.	Intra	–	No	M2	Core	Dynamic algorithm selection among lightweight ciphers on ESP-32; energy/robustness/latency measured. Decision logic (AI vs rules) not recorded.
A37	[37]	Y	P	Closed	Yes	Intra	–	No	M1	Core	Adaptive ABE with ML/context in smart IoT; latency/resource/resilience measured.
A38	[38]	P	P	Closed	n.r.	Inter	PQC	No	M1	Core	Cryptographic agility with adaptive lattice-based PQC for blockchain; energy/efficiency measured. Decision logic not recorded.
A39	[39]	Y	P	Closed	Yes	U	Unspec.	No	M1	Core	DRL + FL with 'quantum-blockchain' for 6G IoMT. 'Quantum' component unspecified in extraction: not verifiable as PQC or QKD -> not quantum-ready.
A40	[40]	P	P	Closed	Yes	Intra	QKD	QKD-assisted	M1	Core	DRL-optimised QKD with VAE anomaly detection; key rate/QBER measured; quantum-network context.
A41	[41]	N	P	Closed	Yes	Intra	Q-inspired	No	M1	Contextual	Adaptive encryption + dynamic key generation for big data; 'quantum-chaotic' = chaotic-map (quantum-inspired) cryptography, not quantum cryptography; not a smart environment.
A42	[42]	Y	Y	Closed	n.r.	Intra	–	No	M1	Core	Adaptive edge security policy framework for IoT; adaptability/resilience assessed. Decision logic not recorded.
A43	[43]	Y	N	Partial	Det.-only	None	–	No	M1	Contextual	Tri-Fusion Net + PLBO for SCADA attack detection; detection-only.
A44	[44]	N	N	Partial	Det.-only	None	–	No	M1	Background	Ensemble NIDS with adversarial alignment for enterprise networks; detection-only, no smart environment.
A45	[45]	Y	N	Partial	Det.-only	None	–	No	M1	Contextual	DNN clone-ID attack detection in RPL IoT; no identified PQC mechanism; detection-only.
A46	[46]	P	P	Closed	Yes	Hybrid	QKD	QKD-assisted	M1	Core	DRL + moving-target defence for network slicing with QKD; fallback to classical/PQC modes reported in text.
A47	[47]	Y	Y	Closed	Rule/opt.	Intra	–	No	M1	Core	Multi-layered IoMT architecture with dynamic key management and decentralised IDS.
A48	[48]	Y	U	Partial	Det.-only	U	Unspec.	No	M1	Contextual	'Quantum-driven' AI-powered cryptography and anomaly detection for IoT; detection metrics only. Quantum mechanism unspecified.
A49	[49]	Y	N	None	No	None	–	No	M0	Background	Secondary study (survey of QoS-based IoT cybersecurity) – conflicts with IC1; retained as background only.
A50	[50]	Y	P	Closed	Yes	U	Unspec.	No	M2	Core	Learning-informed trust controller with approve/throttle/quarantine actions; Raspberry Pi and Jetson Nano evaluation (M2); no identified PQC primitive.
A51	[51]	Y	N	None	No	None	QML/QAI	No	M0	Background	Conceptual AI + quantum 'holographic' framework for IoT; no mechanism or evaluation.
A52	[52]	N	N	Partial	Det.-only	None	–	No	M1	Background	Signature-based IDS with ML/DL and fuzzy clustering; generic IDS, detection-only.
A53	[53]	N	N	None	Yes	None	PQC	No	M1	Background	GA-GAN 'quantum-resistant' key/cipher generation. Not a standardised PQC family; quantum-resistance claim heuristic -> reclassified as 'claimed quantum-resistant (non-standard)'.
A54	[54]	Y	P	Partial	Det.-only	Inter	PQC	No	M1	Contextual	Blockchain + PQC with GNN threat detection for smart agriculture; PQC μsed as drop-in; no runtime switching evaluated.
A55	[55]	N	N	None	No	None	Unspec.	No	M0	Background	ANN-ISM cybersecurity framework for software development; 'quantum' conceptual only.
A56	[56]	Y	P	Closed	Yes	Intra	–	No	M1	Core	Adaptive encryption with ML for fog computing; efficiency/security improvements measured.
A57	[57]	Y	N	None	No	None	–	No	M1	Contextual	DNA-based lightweight cryptography for IoT; single static scheme; timing/resource measured.
A58	[58]	P	N	None	No	None	–	No	M1	Background	Hybrid ECC-AES for cloud data protection; static scheme; no smart-environment context.
A59	[59]	Y	N	Partial	Det.-only	None	–	No	M1	Contextual	DNN + blockchain anomaly detection/prevention for IoT; detection metrics.
A60	[60]	P	P	Closed	Rule/opt.	Intra	–	No	M1	Core	DRAFTEE self-adaptive BFT consensus trading throughput against security constraints.
A61	[61]	Y	P	Closed	Yes	U	–	No	M1	Core	DRL-based self-adaptive intrusion detection optimising security and energy; no identified PQC mechanism.

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Table S1 – continued

ID	Ref.	SE	MW	Loop	AI	CA	Q. family	Q-ready	Mat.	Class	Evidence in extraction / caveats
A62	[62]	N	N	None	No	None	PQC	No	M1	Background	Lattice-based puncturable signatures with 'adaptive security' (cryptographic notion); primitive only.
A63	[63]	Y	P	Partial	No	Inter	PQC	Yes	M1	Contextual	QRSecOC automotive PQC; simulation (M1), no real automotive ECU validation; no evaluated runtime switching.
A64	[64]	Y	P	Closed	Rule/opt.	Intra	PQC	Yes	M1	Core	DAME-IoV dynamic adaptive multi-edge authentication with lattice PQC; <50 ms; 1000+ requests.
A65	[65]	Y	P	Closed	Yes	Intra	–	No	M1	Core	Adaptive security manifold with federated meta-learning in 6G cognitive networks.
A66	[66]	Y	P	Closed	Rule/opt.	Intra	–	No	M1	Core	ASID adaptive physical security for IIoMT with blockchain integrity (PQC only as future work); -85% tampering risk.
A67	[67]	Y	P	Partial	Rule/opt.	None	–	No	M1	Contextual	Adaptive differential evolution + blockchain comms for drone swarms; optimisation of operations, not security adaptation.
A68	[68]	Y	Y	Partial	No	Inter	PQC	No	M0	Contextual	One study linking SSRN discovery and journal publication; journal version retained within the search window according to the author; contextual, M0, no counted quantum-readiness.
A69	[69]	Y	N	None	No	None	PQC	No	M0	Background	Secondary study (review of quantum-secure IoT/6G) – conflicts with IC1; retained as background only.
A70	[70]	Y	Y	Closed	Rule/opt.	Inter	PQC	Yes	M2	Core	Rule-based context-aware key management with lattice PQC on microcontrollers; laboratory prototype (M2), not field deployment.
A71	[71]	P	N	None	No	None	PQC	No	M0	Background	Conceptual analysis of threats/challenges of digital-twin-enabled self-adaptive systems (book chapter).
A72	[72]	P	Y	Closed	Rule/opt.	Intra	–	No	M3	Core	SAFER-D rule-based security adaptation with state transitions and heartbeat coordination; multi-node cloud/fog/edge testbed (M3).
A73	[73]	N	N	None	No	None	QML/QAI	No	M1	Background	Hybrid quantum sparse coding + capsule network for image classification; unrelated to security adaptation.
A74	[74]	P	N	None	No	None	QML/QAI	No	M0	Background	Conceptual perspective on quantum AI and cryptographic security; no mechanism or evaluation.

2 Table S2: Study-level quality-appraisal scores

Study-level quality-appraisal scores (rubric of Table 13 of the main article; scale 0 = absent, 1 = partial, 2 = full). RD = research design; AC = architectural clarity; VE = validation environment; MS = measurements; BL = baselines; TM = threat model; RP = reproducibility; ST = statistical treatment; LM = limitations. The author-confirmed appraisal supplies nine scores for every study (666 cells), with totals out of 18 and no unscored criteria. Scores qualify interpretation rather than eligibility; the rubric does not provide a validated overall risk-of-bias instrument.

Table S2. Study-level quality-appraisal scores.

ID	Ref.	RD	AC	VE	MS	BL	TM	RP	ST	LM	Score	Unscored crit.
A01	[1]	2	1	1	2	1	1	1	1	1	11/18	0
A02	[2]	2	1	1	2	1	1	1	1	1	11/18	0
A03	[3]	2	1	1	2	1	1	1	1	1	11/18	0
A04	[4]	2	1	1	2	1	1	1	1	1	11/18	0
A05	[5]	1	1	0	0	0	1	0	0	1	4/18	0
A06	[6]	2	1	1	2	1	1	1	1	1	11/18	0
A07	[7]	2	1	1	2	1	1	1	1	1	11/18	0
A08	[8]	2	1	1	2	1	1	1	1	1	11/18	0
A09	[9]	2	1	1	2	1	1	1	1	1	11/18	0
A10	[10]	2	1	1	2	1	1	1	1	1	11/18	0
A11	[11]	2	1	1	2	1	1	1	1	1	11/18	0
A12	[12]	2	1	1	2	1	1	1	1	1	11/18	0
A13	[13]	2	1	1	2	1	1	1	1	1	11/18	0
A14	[14]	1	1	0	0	0	1	0	0	1	4/18	0
A15	[15]	1	1	0	0	0	1	0	0	1	4/18	0
A16	[16]	1	1	0	0	0	1	0	0	1	4/18	0
A17	[17]	2	1	1	2	1	1	1	1	1	11/18	0
A18	[18]	2	1	1	2	1	1	1	1	1	11/18	0
A19	[19]	2	1	1	2	1	1	1	1	1	11/18	0

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Table S2 – continued

ID	Ref.	RD	AC	VE	MS	BL	TM	RP	ST	LM	Score	Unscored crit.
A20	[20]	2	2	2	2	1	1	1	1	2	14/18	0
A21	[21]	2	1	1	2	1	1	1	1	1	11/18	0
A22	[22]	2	1	1	2	1	1	1	1	1	11/18	0
A23	[23]	2	1	1	2	1	1	1	1	1	11/18	0
A24	[24]	2	1	1	2	1	1	1	1	1	11/18	0
A25	[25]	2	1	1	2	1	1	1	1	1	11/18	0
A26	[26]	2	1	1	2	1	1	1	1	1	11/18	0
A27	[27]	2	1	1	2	1	1	1	1	1	11/18	0
A28	[28]	1	1	0	0	0	1	0	0	1	4/18	0
A29	[29]	2	1	1	2	1	1	1	1	1	11/18	0
A30	[30]	1	1	0	0	0	1	0	0	1	4/18	0
A31	[31]	2	1	1	2	1	1	1	1	1	11/18	0
A32	[32]	2	1	1	2	1	1	1	1	1	11/18	0
A33	[33]	2	1	1	2	1	1	1	1	1	11/18	0
A34	[34]	2	2	2	2	1	1	1	1	2	14/18	0
A35	[35]	1	1	0	0	0	1	0	0	1	4/18	0
A36	[36]	2	2	2	2	1	1	1	1	2	14/18	0
A37	[37]	2	1	1	2	1	1	1	1	1	11/18	0
A38	[38]	2	1	1	2	1	1	1	1	1	11/18	0
A39	[39]	2	1	1	2	1	1	1	1	1	11/18	0
A40	[40]	2	1	1	2	1	1	1	1	1	11/18	0
A41	[41]	2	1	1	2	1	1	1	1	1	11/18	0
A42	[42]	2	1	1	2	1	1	1	1	1	11/18	0
A43	[43]	2	1	1	2	1	1	1	1	1	11/18	0
A44	[44]	2	1	1	2	1	1	1	1	1	11/18	0
A45	[45]	2	1	1	2	1	1	1	1	1	11/18	0
A46	[46]	2	1	1	2	1	1	1	1	1	11/18	0
A47	[47]	2	1	1	2	1	1	1	1	1	11/18	0
A48	[48]	2	1	1	2	1	1	1	1	1	11/18	0
A49	[49]	1	1	0	0	0	1	0	0	1	4/18	0
A50	[50]	2	2	2	2	1	1	1	1	2	14/18	0
A51	[51]	1	1	0	0	0	1	0	0	1	4/18	0
A52	[52]	2	1	1	2	1	1	1	1	1	11/18	0
A53	[53]	2	1	1	2	1	1	1	1	1	11/18	0
A54	[54]	2	1	1	2	1	1	1	1	1	11/18	0
A55	[55]	1	1	0	0	0	1	0	0	1	4/18	0
A56	[56]	2	1	1	2	1	1	1	1	1	11/18	0
A57	[57]	2	1	1	2	1	1	1	1	1	11/18	0
A58	[58]	2	1	1	2	1	1	1	1	1	11/18	0
A59	[59]	2	1	1	2	1	1	1	1	1	11/18	0
A60	[60]	2	1	1	2	1	1	1	1	1	11/18	0
A61	[61]	2	1	1	2	1	1	1	1	1	11/18	0
A62	[62]	2	1	1	2	1	1	1	1	1	11/18	0
A63	[63]	2	1	1	2	1	1	1	1	1	11/18	0
A64	[64]	2	1	1	2	1	1	1	1	1	11/18	0
A65	[65]	2	1	1	2	1	1	1	1	1	11/18	0
A66	[66]	2	1	1	2	1	1	1	1	1	11/18	0
A67	[67]	2	1	1	2	1	1	1	1	1	11/18	0
A68	[68]	1	1	0	0	0	1	0	0	1	4/18	0
A69	[69]	1	1	0	0	0	1	0	0	1	4/18	0
A70	[70]	2	2	2	2	1	1	1	1	2	14/18	0
A71	[71]	1	1	0	0	0	1	0	0	1	4/18	0
A72	[72]	2	2	2	2	2	2	1	2	2	17/18	0
A73	[73]	2	1	1	2	1	1	1	1	1	11/18	0
A74	[74]	1	1	0	0	0	1	0	0	1	4/18	0

3 Table S3: Taxonomy index

Taxonomy index: study identifiers attached to each category of the four taxonomy figures (Figures 3–6 of the main article). Identifiers refer to Appendix A of the main article; a study may appear under several categories. n is the number of distinct studies; C/X/B gives how many of them are core, contextual and background studies under the scope classes of Table S1. Mappings are given at leaf level where the figure annotation is unambiguous and at category level (“all leaves”) otherwise; “!” marks categories whose leaf annotation in the figure is aggregated, so the mapping is given at category level. The last row of each block lists the studies that do not appear in that figure.

Table S3. Taxonomy categories and supporting studies.

Fig.	Category	Leaf	Study identifiers	*n*	C/X/B
FQ1	With AI: Reinforcement learning	Deep Q-networks	A18, A39, A40, A65	4	4/0/0
FQ1	With AI: Reinforcement learning	Policy-gradient	A39, A46	2	2/0/0
FQ1	With AI: Reinforcement learning	Q-learning	A11, A26, A61	3	2/1/0
FQ1	With AI: Supervised learning	LDA	A25, A33, A43, A52	4	0/2/2
FQ1	With AI: Supervised learning	Naive Bayes	A25, A33, A52	3	0/1/2
FQ1	With AI: Supervised learning	Logistic regression	A25, A33, A34, A52	4	0/2/2
FQ1	With AI: Supervised learning	Gradient boosted trees	A33, A44, A52	3	0/0/3
FQ1	With AI: Supervised learning	Random forest	A25, A33, A34, A44, A52	5	0/2/3
FQ1	With AI: Supervised learning	Support vector machines	A25, A34, A45, A52	4	0/3/1
FQ1	With AI: Supervised learning	k-nearest neighbours	A25, A34, A45	3	0/3/0
FQ1	With AI: Unsupervised learning	Clustering	A41, A52	2	0/1/1
FQ1	With AI: Unsupervised learning	Variational autoencoder	A40, A41	2	1/1/0
FQ1	With AI: Deep learning architectures	Convolutional neural networks	A25, A34, A39, A50	4	2/2/0
FQ1	With AI: Deep learning architectures	Recurrent networks	A24, A39, A50	3	2/1/0
FQ1	With AI: Deep learning architectures	LSTM	A24	1	0/1/0
FQ1	With AI: Deep learning architectures	Hybrid deep architectures	A39, A48, A50	3	2/1/0
FQ1	With AI: Deep learning architectures	Transformers	A39, A48	2	1/1/0
FQ1	With AI: Distributed learning	Federated learning	A18, A19, A28, A39, A61, A65	6	5/1/0
FQ1	With AI: Distributed learning	Meta-learning	A65	1	1/0/0
FQ1	With AI: Probabilistic	Bayesian trust models	A07, A32, A39, A47	4	4/0/0
FQ1	With AI: Probabilistic	Learning-based reputation	A32, A39, A47, A49, A66	5	4/0/1
FQ1	Knowledge and telemetry support	Ontology repositories	A07, A20, A22, A32, A37, A42, A66, A70, A72	9	9/0/0
FQ1	Knowledge and telemetry support	Telemetry pipelines	A24, A33, A41, A49, A60, A70, A73	7	2/2/3
FQ1	Knowledge and telemetry support	Blockchain	A32, A39, A54, A59, A66, A67, A68, A69	8	3/4/1
FQ1	Knowledge and telemetry support	Quantum telemetry	A39, A40, A46, A63, A64, A70, A71	7	5/1/1
FQ1	Without AI: Rule-based engines	Context-aware rule engines	A08, A10, A12, A17, A20, A22, A31, A37, A42, A66, A70, A72	12	9/1/2
FQ1	Without AI: Rule-based engines	Risk matrices	A02, A49, A66	3	2/0/1
FQ1	Without AI: Fuzzy logic controllers	Fuzzy inference systems	A04, A07, A52	3	2/0/1
FQ1	Without AI: Game-theoretic models	Game-theoretic security policies	A02	1	1/0/0
FQ1	Without AI: Tree-based security models	Attack–defence trees	A27, A43, A60	3	2/1/0

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Fig.	Category	Leaf	Study identifiers	*n*	C/X/B
FQ1	Without AI: Metaheuristic	Multi-objective optimisation	A39, A48, A50	3	2/1/0
FQ1	Without AI: Metaheuristic	Differential evolution	A67	1	0/1/0
FQ1	Without AI: Metaheuristic	Particle swarm optimisation	A01, A06, A23, A36, A67, A70	6	3/1/2
FQ1	Without AI: Metaheuristic	Genetic algorithms	A01, A13, A23, A36, A67	5	1/1/3
FQ1	Without AI: KPI-based	Performance KPI thresholds	A03, A06, A23, A24, A49, A60, A63, A64, A70	9	4/2/3
FQ1	Without AI: KPI-based	Quantum KPI thresholds	A39, A40, A46, A63, A64, A70, A71	7	5/1/1
FQ1	Without AI: Conceptual configuration	Static cryptographic profiles	A09, A14, A15, A16, A30, A38, A51, A53, A55, A57, A58, A62, A71, A73, A74	15	1/3/11
FQ1 (Figure 3): 69 of 74 studies appear; not shown: A05, A21, A29, A35, A56.					
FQ2	Performance: Network performance	(all leaves)	A03, A04, A06, A07, A08, A18, A19, A20, A23, A24, A25, A27, A31, A32, A40, A42, A49, A60, A63, A64, A65, A70	22	15/4/3
FQ2	Performance: Computational performance	(all leaves)	A06, A08, A09, A12, A13, A17, A18, A20, A21, A23, A24, A27, A30, A31, A36, A38, A45, A49, A57, A58, A60, A61, A62, A63, A64, A65, A70	27	13/6/8
FQ2	Performance: Resource efficiency	(all leaves)	A06, A18, A20, A24, A26, A31, A32, A36, A37, A38, A42, A45, A49, A56, A57, A58, A60, A61, A65, A67, A70	21	14/5/2
FQ2	Performance: System reliability	(all leaves)	A03, A23, A27, A33, A40, A42, A43, A44, A47, A49, A60, A66	12	6/1/5
FQ2	Intrusion-detection metrics	(all leaves)	A11, A20, A24, A25, A26, A29, A33, A34, A40, A41, A43, A44, A45, A47, A50, A52, A54, A56, A59, A61	20	7/10/3
FQ2	Cryptographic performance	(all leaves)	A20, A30, A38, A41, A57, A58, A62, A63, A68	9	2/4/3
FQ2	Quantum-cryptography metrics	(all leaves)	A39, A40, A46, A63, A64, A70, A71	7	5/1/1
FQ2	Adaptation and resilience metrics	(all leaves)	A08, A18, A19, A20, A22, A26, A27, A36, A37, A39, A42, A46, A60, A61, A65, A67, A70	17	16/1/0
FQ2	Security: Authentication	(all leaves)	A06, A08, A10, A12, A17, A20, A31, A32, A37, A63, A64, A66, A70	13	9/2/2
FQ2	Security: Attack and threat metrics	(all leaves)	A11, A20, A22, A25, A26, A27, A29, A33, A34, A39, A40, A41, A42, A43, A44, A47, A52, A54, A61, A66	20	10/7/3
FQ2	Security: Reputation metrics	(all leaves)	A07, A32, A39, A47, A49, A66, A68	7	5/1/1
FQ2	Security: Data integrity	(all leaves)	A12, A17, A19, A20, A22, A28, A31, A33, A42, A47, A66, A67	12	6/3/3
FQ2	Contextual factors: Operational context !	(all leaves)	A02, A03, A06, A07, A08, A18, A20, A22, A23, A24, A27, A42, A49, A60, A63, A64, A65, A66, A70	19	14/2/3
FQ2	Contextual factors: Role context	(all leaves)	A02, A07, A08, A10, A12, A17, A20, A22, A31, A32, A37, A39, A42, A47, A49, A66, A70, A72	18	14/1/3
FQ2	Contextual factors: Data sensitivity	(all leaves)	A02, A07, A08, A10, A12, A17, A18, A20, A22, A24, A27, A31, A35, A37, A39, A42, A47, A66, A71, A73	20	13/2/5
FQ2	Contextual factors: Compliance	(all leaves)	A02, A07, A08, A12, A17, A20, A31, A39, A42, A47, A66, A70	12	9/1/2
FQ2	Contextual factors: Risk context	(all leaves)	A02, A07, A11, A16, A19, A20, A25, A26, A27, A28, A29, A33, A34, A39, A40, A41, A42, A43, A44, A45, A46, A47, A48, A49, A50, A52, A54, A56, A57, A59, A61, A64, A66, A70, A72	35	18/13/4
FQ2	Contextual factors: Device and network context !	(all leaves)	A02, A04, A06, A07, A18, A19, A20, A22, A24, A25, A26, A30, A32, A36, A39, A42, A45, A46, A47, A49, A57, A61, A63, A64, A65, A66, A70, A74	28	20/5/3
FQ2 (Figure 4): 66 of 74 studies appear; not shown: A01, A05, A14, A15, A51, A53, A55, A69.					

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Fig.	Category	Leaf	Study identifiers	*n*	C/X/B
FQ3	Health and IoMT	Privacy, availability and latency	A02, A07, A39, A47, A50, A66	6	6/0/0
FQ3	Transport and vehicular networks	Mobility, authentication and deadlines	A04, A25, A63, A64, A68	5	2/3/0
FQ3	Smart homes	Energy, heterogeneity and privacy	A22, A36, A37	3	3/0/0
FQ3	Industry, energy and cyber-physical infrastructure	Safety, availability and determinism	A03, A20, A27, A31, A43, A71, A72	7	3/2/2
FQ3	Agriculture	Energy, connectivity and remote operation	A54	1	0/1/0
FQ3	Resource-constrained sensors and IoT	Memory, energy and cryptographic cost	A10, A26, A45, A57, A61, A70	6	4/2/0
FQ3	UAV and aerospace systems	Mobility, intermittent links and energy	A18, A24, A67	3	1/2/0
FQ3	Generic IoT, edge and fog	Coordination, heterogeneity and distributed trust	A08, A19, A28, A29, A32, A42, A48, A49, A56, A59	10	5/4/1
FQ3	Networks and distributed systems	Scalability, coordination and convergence	A06, A21, A38, A60, A65	5	5/0/0
FQ3	QKD networks	Key rate, QBER and fallback	A40, A46	2	2/0/0
FQ3	Surveillance and connected cameras	Latency, privacy and telemetry integrity	A34	1	0/1/0
FQ3	Cloud, storage and generic security	Access, storage and isolation	A12, A14, A16, A17, A33, A44, A52, A58	8	0/1/7
FQ3	No specific smart environment	No deployment constraint assigned	A01, A05, A09, A11, A13, A15, A23, A30, A35, A41, A51, A53, A55, A62, A69, A73, A74	17	0/3/14
FQ3 (Figure 5): 74 of 74 studies appear; not shown: none.					
FQ4	Classical cryptography	Generic classical cryptography	A02, A03, A04, A06, A11, A16, A17, A18, A22, A23, A25, A42, A43, A44, A49, A52, A72, A74	18	7/4/7
FQ4	Classical cryptography	Lightweight symmetric encryption	A07, A36, A56, A70	4	4/0/0
FQ4	Classical cryptography	Classical digital signatures	A04	1	1/0/0
FQ4	Classical cryptography	DNA-based lightweight cryptography	A57	1	0/1/0
FQ4	Classical cryptography	ECC and hybrid ECC–AES encryption	A58	1	0/0/1
FQ4	Classical cryptography	Network-coding-based cryptography	A14	1	0/0/1
FQ4	Post-quantum cryptography	Lattice-based post-quantum cryptography	A05, A09, A38, A62, A64, A70	6	3/1/2
FQ4	Post-quantum cryptography	Generic post-quantum cryptography	A20, A30, A53, A63, A69, A71	6	1/1/4
FQ4	Post-quantum cryptography	Post-quantum signcryption	A68	1	0/1/0
FQ4	Quantum and quantum-inspired (superseded, see Table 26 of the main article)	Quantum key distribution	A40, A46	2	2/0/0
FQ4	Quantum and quantum-inspired (superseded, see Table 26 of the main article)	Quantum-chaotic cryptography	A39, A41, A48, A51, A55	5	1/2/2
FQ4	Quantum and quantum-inspired (superseded, see Table 26 of the main article)	Quantum-inspired metaheuristic	A01, A15	2	0/0/2

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Fig.	Category	Leaf	Study identifiers	*n*	C/X/B
FQ4	Key management and rotation	Fine-grained access control	A08, A10, A31, A37	4	3/1/0
FQ4	Key management and rotation	Chaos-based	A13, A41	2	0/1/1
FQ4	Key management and rotation	Classical key management and key rotation	A21, A26, A47	3	3/0/0
FQ4	Key management and rotation	Context-aware key management with PQC	A70	1	1/0/0
FQ4	Blockchain-centric	Blockchain with classical cryptography	A39, A41, A54, A59, A60, A67	6	2/4/0
FQ4	Blockchain-centric	Blockchain with post-quantum cryptography	A54, A66, A68, A69, A70	5	2/2/1
FQ4	Blockchain-centric	Blockchain with quantum-chaotic components	A39, A41	2	1/1/0
FQ4	Security architectures	Self-adaptive security and trust frameworks	A27, A31, A65, A71, A72, A74	6	3/1/2
FQ4	Security architectures	Decentralised trust architectures	A19, A32, A39	3	3/0/0

FQ4 (Figure 6): 63 of 74 studies appear; not shown: A12, A24, A28, A29, A33, A34, A35, A45, A50, A61, A73.

References

[1] Qiuyi Wu, Licheng Jiao, Yangyang Li, and Xiaozheng Deng. A novel quantum-inspired immune clonal algorithm with the evolutionary game approach. *Progress in Natural Science*, 19(10):1341–1347, 2009. doi: 10.1016/j.pnsc.2009.02.007.

[2] Habtamu Abie and Ilangko Balasingham. Risk-based adaptive security for smart iot in ehealth. In *Proceedings of the 7th International Conference on Body Area Networks*, page 269–275. ICST, 2012. doi: 10.5555/2442691.2442752.

[3] H. Shareef, A.A. Ibrahim, N. Salman, A. Mohamed, and W. Ling Ai. Power quality and reliability enhancement in distribution systems via optimum network reconfiguration by using quantum firefly algorithm. *International Journal of Electrical Power & Energy Systems*, 58:160–169, 2014. doi: 10.1016/j.ijepes.2014.01.013.

[4] Muhammad Awais Javed, Sherali Zeadally, Muhammad Usman, and Guftaar Ahmad Sardar Sidhu. Faspm: Fuzzy logic-based adaptive security protocol for multihop data dissemination in intelligent transport systems. *Transactions on Emerging Telecommunications Technologies*, 28(11):e3190, 2017. doi: 10.1002/ett.3190.

[5] Olivier Blazy, Céline Chevalier, and Quoc Huy Vu. Post-quantum uc-secure oblivious transfer in the standard model with adaptive corruptions. In *Proceedings of the 14th International Conference on Availability, Reliability and Security*. Association for Computing Machinery, 2019. doi: 10.1145/3339252.3339280.

[6] Bomin Mao, Yuichi Kawamoto, and Nei Kato. Ai-based joint optimization of qos and security for 6g energy harvesting internet of things. *IEEE Internet of Things Journal*, 7(8):7032–7042, 2020. doi: 10.1109/JIOT.2020.2982417.

[7] Tidiane Sylla, Mohamed Aymen Chalouf, Francine Krief, and Karim Samaké. Setucom: Secure and trustworthy context management for context-aware security and privacy in the internet of things. *Security and Communication Networks*, 2021(1):6632747, 2021. doi: 10.1155/2021/6632747.

[8] Timam Ghosh, Arijit Roy, Sudip Misra, and Narendra Singh Raghuwanshi. Case: A context-aware security scheme for preserving data privacy in iot-enabled society 5.0. *IEEE Internet of Things Journal*, 9(4):2497–2504, 2022. doi: 10.1109/JIOT.2021.3101115.

[9] Dharminder Dharminder, Ashok Kumar Das, Sourav Saha, Basudeb Bera, and Athanasios V. Vasilakos. Post-quantum secure identity-based encryption scheme using random integer lattices for iot-enabled ai applications. *Security and Communication Networks*, 2022(1):5498058, 2022. doi: 10.1155/2022/5498058.

[10] Saad Inshi, Rasel Chowdhury, Hakima Ould-Slimane, and Chamseddine Talhi. Dynamic context-aware security in a tactical network using attribute-based encryption. In *MILCOM 2022 - 2022 IEEE Military Communications Conference (MILCOM)*, pages 49–54, 2022. doi: 10.1109/MILCOM55135.2022.10017647.

[11] Sereysethy Touch and Jean-Noël Colin. A comparison of an adaptive self-guarded honeypot with conventional honeypots. *Applied Sciences*, 12(10):5224, 2022. doi: 10.3390/app12105224.

[12] Guowen Xu, Shengmin Xu, Jinhua Ma, Jianting Ning, and Xinyi Huang. An adaptively secure and efficient data sharing system for dynamic user groups in cloud. *IEEE Transactions on Information Forensics and Security*, 18:5171–5185, 2023. doi: 10.1109/TIFS.2023.3305870.

[13] Nawal Shaltout, Ahmed A.Abd El-Latif, Waleed M. El-Latif, and Samir Elmougy. Applicable image security based on computational genetic approach and self-adaptive substitution. *IEEE Access*, 11:2303–2317, 2023. doi: 10.1109/ACCESS.2022.3233321.

[14] Naveed N, Nanda Kishor K P, and Niranjan R T. The adaptive security of cloud information management encrypted with cryptographic network coding. In *2023 International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering (ICECONF)*, pages 1–7, 2023.

doi: 10.1109/ICECONF57129.2023.10083532.

[15] Hui Shao and Anders W. Sandvik. Progress on stochastic analytic continuation of quantum monte carlo data. *Physics Reports*, 1003:1–88, 2023. doi: 10.1016/j.physrep.2022.11.002.

[16] Aakash Ahmad, Asad Waqar Malik, Abdulrahman Alreshidi, Wilayat Khan, and Maryam Sajjad. Adaptive security for self-protection of mobile computing devices. *Mobile Networks and Applications*, 28(2):653–672, 2023. doi: 10.1007/s11036-019-01355-y.

[17] Shengmin Xu, Xingshuo Han, Guowen Xu, Jianting Ning, Xinyi Huang, and Robert H. Deng. An adaptive secure and practical data sharing system with verifiable outsourced decryption. *IEEE Transactions on Services Computing*, 17(3):776–788, 2024. doi: 10.1109/TSC.2023.3321314.

[18] Uttam Ghosh, Laurent Njilla, Debashis Das, and Eugene Levin. Flas: A federated learning framework for adaptive security in edge-driven uav networks. In *ICC 2024 - IEEE International Conference on Communications*, pages 1298–1303, 2024. doi: 10.1109/ICC51166.2024.10622352.

[19] Paul Shepherd, Tasos Dagiuklas, Bugra Alkan, and Jonathan Rodriguez. Integrating a zero-trust adaptive security framework in edge based federated learning. In *2024 IEEE 29th International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD)*, pages 1–5, 2024. doi: 10.1109/CAMAD62243.2024.10942760.

[20] Arturo Holgado, Jorge Portilla, Daniel López-Fernández, and Luis Redondo. Context-aware security and post quantum cryptography applied to iot networks. In *2024 4th Intelligent Cybersecurity Conference (ICSC)*, pages 175–179, 2024. doi: 10.1109/ICSC63108.2024.10894874.

[21] Lyubomir Nenov. Reinforcement learning for key management in distributed systems. In *2024 32nd National Conference with International Participation (TELECOM)*, pages 1–5, 2024. doi: 10.1109/TELECOM63374.2024.10812245.

[22] Josival Silva, Nelson Rosa, and Fernando Aires. Up-home: A self-adaptive solution for smart home security. *JUCS - Journal of Universal Computer Science*, 30(4):502–530, 2024. doi: 10.3897/jucs.107050.

[23] Yuhang Lin, Limei Lin, Yanze Huang, Li Xu, and Sun-Yuan Hsieh. Endogenous security of fq_n networks: Adaptive system-level fault self-diagnosis. *IEEE Transactions on Reliability*, 73(3):1659–1668, 2024. doi: 10.1109/TR.2024.3356200.

[24] Syed Zubair Ahmad and Farhan Qamar. A hybrid ai based framework for enhancing security in satellite based iot networks using high performance computing architecture. *Scientific Reports*, 14(1):30695, 2024. doi: 10.1038/s41598-024-78262-0.

[25] Thiruppathy Kesavan Venkatasamy, Md. Jakir Hossen, Gopi Ramasamy, and Nor Hidayati Binti Abdul Aziz. Intrusion detection system for v2x communication in vanet networks using machine learning-based cryptographic protocols. *Scientific Reports*, 14(1):31780, 2024. doi: 10.1038/s41598-024-82313-x.

[26] Xiaofen Fang, Lihui Zheng, Xiaohua Fang, Weidong Chen, Kunli Fang, Lingpeng Yin, and Han Zhu. Pioneering advanced security solutions for reinforcement learning-based adaptive key rotation in zigbee networks. *Scientific Reports*, 14(1):13931, 2024. doi: 10.1038/s41598-024-64895-8.

[27] Salim Chehida, Eric Rutten, Guillaume Giraud, and Stéphane Mocanu. A model-based approach for self-adaptive security in cps: Application to smart grids. *Journal of Systems Architecture*, 150:103118, 2024. doi: 10.1016/j.sysarc.2024.103118.

[28] Aziz Ullah Karimy and P. Chandrasekhar Reddy. Analyzing federated learning as a novel approach for enhancing security and privacy in the internet of things (iot). In *2024 Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, pages 1–7, 2024. doi: 10.1109/ICAECT60202.2024.10468686.

[29] Venkata Sai Swaroop Reddy Nallapareddy. Machine learning-based adaptive cybersecurity framework for the internet of things. In *2024 International Conference on Intelligent Computing and Emerging Communication Technologies (ICEC)*, pages 1–6, 2024. doi: 10.1109/ICEC59683.2024.10837268.

[30] Muhammad Nauman Khan. *Lightweight and post-quantum safe security solutions for IoT systems*. Thesis, RMIT University, 2024.

[31] Minghao Yang and Hu Xiong. Efficient and decentralized dual access control for cloud-based industrial internet of things. *IEEE Internet of Things Journal*, 12(12):21082–21090, 2025. doi: 10.1109/JIOT.2025.3547044.

[32] Ijaz Ahmad, Shakthi Gimhana, Ijaz Ahmad, and Erkki Harjula. Adaptive trust architecture for secure iot communication in 6g. *IEEE Networking Letters*, 7(2):113–116, 2025. doi: 10.1109/LNET.2025.3566909.

[33] Ahmed Saad Alsalim. Improving the security of enterprise storage systems with efficient anomaly detection. *IEEE Reliability Magazine*, 2(2):35–39, 2025. doi: 10.1109/MRL.2025.3556806.

[34] Anakhi Hazarika, Nikumani Choudhury, Lei Shu, and Qin Su. Real-time detection of cyber-physical attacks on smart-ip-camera using network and telemetry data. *IEEE Transactions on Industrial Cyber-Physical Systems*, 3:251–261, 2025. doi: 10.1109/TICPS.2025.3544128.

[35] Zahir Tari and Redowan Mahmud. Augmenting digital ecosystem resilience through human-centric cybersecurity solutions. *IEEE Transactions on Engineering Management*, 72:3892–3908, 2025. doi: 10.1109/TEM.2025.3606637.

[36] Shihab Hossen Rafat, Mishat Nissat Jarin, Taher Muhammad Mahdee, Abu Saleh Musa Miah, Najmul Haque, Md. Mahbubur Rahman Tusher, Jungpil Shin, and Istiaque Ahmed. Lightweight cryptographic algorithm analysis for secure iot communication on esp-32 platforms. In *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, pages 1–6, 2025. doi: 10.1109/QPAIN66474.2025.11171742.

[37] J. Joshua Daniel Raj, Sasikala. T, Ezhilarasan. K, Rajesh. K, Karthik. P, and J. Samson Immanuel. Adaptive attribute-based encryption(a-abe) framework for securing smart iot networks. In *2025 5th International Conference on Soft Computing for Security Applications (ICSCSA)*, pages 249–254, 2025. doi: 10.1109/ICSCSA66339.2025.11171299.

[38] Michail-Alexandros Kourtis, George Oikonomakis, Ioannis Koufos, George Xilouris, Ioannis Kefaloukos, Evangelos Markakis, Daniel Karcz, and Marcin Niemiec. Adaptive post-quantum cryptography for blockchain: Enhancing security and energy efficiency with cryptographic agility. In *2025 6th International Conference in Electronic Engineering & Information Technology (EEITE)*, pages 1–5, 2025. doi: 10.1109/EEITE65381.2025.11166257.

[39] Doha Ar-Reyouchi, Houssam Benali, Benbrik Jihane, Kamal Ghoumid, and El Miloud Ar-Reyouchi. Secure-by-design iomt networks: A quantum-blockchain and ai framework for 6g healthcare. In *2025 International Conference on Communication, Computing, Networking, and Control in*

Cyber-Physical Systems (CCNCPS), pages 244–249, 2025. doi: 10.1109/CCNCPS66785.2025.11135745.

[40] Harshala Shingne, Diptee Chikmurge, Priya Parkhi, and Poorva Agrawal. Design of an integrated model using deep reinforcement learning and variational autoencoders for enhanced quantum security. *MethodsX*, 15:103445, 2025. doi: 10.1016/j.mex.2025.103445.

[41] Archana Kotangale and Meesala Sudhir Kumar. An iterative quantum chaotic security method design for big data deployments using a multi-model framework for key generation, threat modelling, and anomaly-resilient validation. *MethodsX*, 15:103643, 2025. doi: 10.1016/j.mex.2025.103643.

[42] Malka N. Halgamuge and Dusit Niyato. Adaptive edge security framework for dynamic iot security policies in diverse environments. *Computers & Security*, 148:104128, 2025. doi: 10.1016/j.cose.2024.104128.

[43] Alaa O. Khadidos, Adil O. Khadidos, Shitharth Selvarajan, Taher Al-Shehari, Nasser A Alsadhan, and Subhav Singh. Cybersentry: Enhancing scada security through advanced deep learning and optimization strategies. *International Journal of Critical Infrastructure Protection*, 50:100782, 2025. doi: 10.1016/j.ijcip.2025.100782.

[44] Khandakar Md Shafin, G. M. Abdullah Al Kafi, and Saha Reno. Guardians of the network: An ensemble learning framework with adversarial alignment for evasive cyber threat detection. *Engineering Reports*, 7(10):e70419, 2025. doi: 10.1002/eng2.70419.

[45] Fatima Al-Quayed, Sana Rauf Awan, Noshina Tariq, Mamoona Humayun, Thanaa S Alnusairi, and Tayyab Rehman. Cid-rpl: Clone id attack detection using deep neural network for rpl-based iot networks. *IET Communications*, 19(1):e70067, 2025. doi: 10.1049/cmu2.70067.

[46] Andreas Andreou, Constandinos X. Mavromoustakis, Evangelos Markakis, Athina Bourdena, and George Mastorakis. Enhancing network slice security with deep reinforcement learning and moving target defense strategies. *Discover Internet of Things*, 5(1):67, 2025. doi: 10.1007/s43926-025-00161-1.

[47] Nikhil Sharma and Prashant Giridhar Shambharkar. Multi-layered security architecture for iomt systems: integrating dynamic key management, decentralized storage, and dependable intrusion detection framework. *International Journal of Machine Learning and Cybernetics*, 16(9):6399–6446, 2025. doi: 10.1007/s13042-025-02628-7.

[48] Hana Mohammed Mujlid and Reem Alshahrani. Quantum-driven security evolution in iot: Ai-powered cryptography and anomaly detection. *The Journal of Supercomputing*, 81(9):1087, 2025. doi: 10.1007/s11227-025-07582-3.

[49] Seyed Salar Sefati, Bahman Arasteh, Simona Halunga, and Octavian Fratu. A comprehensive survey of cybersecurity techniques based on quality of service (qos) on the internet of things (iot). *Cluster Computing*, 28(12):792, 2025. doi: 10.1007/s10586-025-05449-z.

[50] Nithesh Naik, Neha Surendranath, Sai Annamaiah Basava Raju, Chennaiah Madduri, Nagaraju Dasari, Vinod Kumar Shukla, and Vathsala Patil. Hybrid deep learning-enabled framework for enhancing security, data integrity, and operational performance in healthcare internet of things (h-iot) environments. *Scientific Reports*, 15(1):31039, 2025. doi: 10.1038/s41598-025-15292-2.

[51] Abdullah I. A. Alzahrani. Exploring ai and quantum computing synergies in holographic counterpart frameworks for iot security and privacy. *The Journal of Supercomputing*, 81(11):1194, 2025. doi: 10.1007/s11227-025-07682-0.

[52] Usama Ahmed, Mohammad Nazir, Amna Sarwar, Tariq Ali, El-Hadi M. Aggoune, Tariq Shahzad, and Muhammad Adnan Khan. Signature-based intrusion detection using machine learning and deep learning approaches empowered with fuzzy clustering. *Scientific Reports*, 15(1):1726, 2025. doi: 10.1038/s41598-025-85866-7.

[53] Purushottam Singh, Prashant Pranav, and Sandip Dutta. A ga-gan approach for next-generation cryptographic security with a focus on quantum-resistant cryptography. *Discover Computing*, 28(1):82, 2025. doi: 10.1007/s10791-025-09594-2.

[54] Ibraheem M. Alharbi and Najah Kalifah Almazmomi. Ai-optimized blockchain security for smart agriculture using post-quantum cryptography and graph neural network-based threat detection. *Peer-to-Peer Networking and Applications*, 18(5):276, 2025. doi: 10.1007/s12083-025-02062-0.

[55] Habib Ullah Khan, Rafiq Ahmad Khan, Hathal S. Alwageed, Alaa Omran Almagrabi, Sarra Ayouni, and Mohamed Maddeh. Ai-driven cybersecurity framework for software development based on the ann-ism paradigm. *Scientific Reports*, 15(1):13423, 2025. doi: 10.1038/s41598-025-97204-y.

[56] Priyanka Rajan Kumar and Sonia Goel. A secure and efficient encryption system based on adaptive and machine learning for securing data in fog computing. *Scientific Reports*, 15(1):11654, 2025. doi: 10.1038/s41598-025-92245-9.

[57] Sehrish Aqeel, Adnan Shahid Khan, Irshad Ahmed Abbasi, Fahad Algarni, and Daniel Grzonka. Enhancing iot security with a dna-based lightweight cryptography system. *Scientific Reports*, 15(1):13367, 2025. doi: 10.1038/s41598-025-96292-0.

[58] P. Selvi and S. Sakthivel. A hybrid ecc-aes encryption framework for secure and efficient cloud-based data protection. *Scientific Reports*, 15(1):30867, 2025. doi: 10.1038/s41598-025-01315-5.

[59] Sathyabama A R and Jeevaa Katiravan. Enhancing anomaly detection and prevention in internet of things (iot) using deep neural networks and blockchain based cyber security. *Scientific Reports*, 15(1):22369, 2025. doi: 10.1038/s41598-025-04164-4.

[60] Guilain Leduc, Sylvain Kubler, and Jean-Philippe Georges. Draftee: A self-adaptive framework for bft-based consensus to meet fluctuating throughput demands under security constraint. *Computer Networks*, 269:111396, 2025. doi: 10.1016/j.comnet.2025.111396.

[61] Saeid Jamshidi, Ashkan Amirnia, Amin Nikanjam, Kawser Wazed Nafi, Foutse Khomh, and Samira Keivanpour. Self-adaptive cyber defense for sustainable iot: A drl-based ids optimizing security and energy efficiency. *Journal of Network and Computer Applications*, 239:104176, 2025. doi: 10.1016/j.jnca.2025.104176.

[62] Jumei Zhang, Zhenhua Liu, and Dongdong Yao. Fine-grained privately puncturable signatures with adaptive security from lattices. In *Proceedings of the 10th International Conference on Cyber Security and Information Engineering*, page 251–255. Association for Computing Machinery, 2025. doi: 10.1145/3759179.3760358.

[63] Amjad Nsour. Quantum-resilient secure onboard communication (qrsecoc): Integrating post-quantum cryptography for robust automotive network security. *International Journal of Computer Networks and Applications (IJCNA)*, 12(1):27–48, 2025. doi: 10.22247/ijcna/2025/03.

[64] Iftikhar Rasheed and Hala Mostafa. Dame-iov: Dynamic adaptive multi-edge authentication protocol with post-quantum security for internet of vehicles. *Vehicular Communications*, 54:100933, 2025. doi: 10.1016/j.vehcom.2025.100933.

[65] Iftikhar Rasheed, Ukasha Awan, Muhammad Adeel, and Muhammad Usama Tayyab. Enhanced security in 6g cognitive networks: A multi-objective approach against advanced threats. *IEEE Communications Letters*, 29(10):2313–2316, 2025. doi: 10.1109/LCOMM.2025.3593801.

[66] Umar Islam, Mohammed Naif Alatawi, Sulaiman Alamro, Hathal Salamah Alwageed, Hanif Ullah, and Naveed Khan. Enhancing physical security in iiomt environments. *Internet of Things*, 33:101653, 2025. doi: 10.1016/j.iot.2025.101653.

[67] Usama Arshad and Zahid Halim. Secure and optimized drone swarm operations with decentralized adaptive differential evolution. *Computers and Electrical Engineering*, 126:110487, 2025. doi: 10.1016/j.compeleceng.2025.110487.

[68] Junhao Li, Qiang Nong, and Ziyu Liu. A resilient fog-enabled iov architecture: Adaptive post-quantum security framework with generalized signcryption and blockchain-enhanced trust management. *Journal of Network and Computer Applications*, 244:104367, 2025. doi: 10.1016/j.jnca.2025.104367.

[69] Karthik Kumar Vaigandla. Quantum-secure iot networks for the 6g era: Post-quantum cryptography, blockchain integration, and trust architectures - a comprehensive review. *Journal of Sensors, IoT & Health Sciences (JSIHS)*, 3:44–75, 2025. doi: 10.69996/jsihs.2025014.

[70] Haythem Hayouni. Adaptive post-quantum security framework for wireless sensor networks using lightweight cryptography and context-aware key management. *The Journal of Supercomputing*, 81(15):1426, 2025. doi: 10.1007/s11227-025-07916-1.

[71] Rahul Sankalana Gunawardhana. Security threats and challenges of digital twins-enabled self-adaptive systems. In *Rebeca for Actor Analysis in Action*, pages 390–405. Springer Nature Switzerland, 2025. doi: 10.1007/978-3-031-85134-6_17.

[72] Marco Stadler, Michael Vierhauser, Michael Riegler, Daniel Waghubinger, and Johannes Sametinger. *SAFER-D: A Self-adaptive Security Framework for Distributed Computing Architectures*, page 197–213. Springer Nature Switzerland, 2025. doi: 10.1007/978-3-032-02138-0_13.

[73] Zhong Shu. Hybrid quantum sparse coding and dynamic convolution capsule network for enhanced image classification. *Pattern Recognition*, 169:111974, 2026. doi: 10.1016/j.patcog.2025.111974.

[74] Muharrem Tuncay Gençoğlu. Quantum artificial intelligence and cryptographic security from a quantum cryptography perspective. *Discover Quantum Science*, 2:4, 2026. doi: 10.1007/s44464-026-00008-w.